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Unit-2

Problem Solving Method

→ Problem Solving approach to typical AI problems.

According to computer scientist a problem solving is a part of AI. which enclose a no. of techniques such as algorithums, heuristic, to solve a problem. (logic, tehniqua)

→ ☆ Step perform by problem solving Agent

(1) Goal formulation :- It organize the steps or sequence required to formulate one goal out of multiple goals as well as action to achieve that goal.

(2) Problem formulation :- It is most imp step in problem solving which decide what action should be taken to achieve the formulation / formulate goal.

☆ There are five components involve in problem formulation.

(i) Initial stage
(ii) Action

2 marks
imp

(3) Transition model :-

Initial state

1	8	2
3		5
4	6	7

Goal state

1	2	3
4	5	6
7	8	9

shifting are called actions.

1	2	
3	8	5
4	6	7

Change (Transition model)

Intermidiate state

→ It is the description of what each action does.

→ Transition model is the fn. of Result (s, a) that return the state as per the given state.

(4) Goal state

→ It determine the if the given state is a goal state.

(5) Path cost (Path ↓ ✓ best)

(which one is the best algorithm according to the path)

→ It assign a numeric cost to each path that follow the goal.

Step (3:)

(3) Search Algorithum

It identify all the best possible sequence to reach the goal state from the current state.

Step (4) Solutions.

It find the best algorithum out of various algorithum which may be known as the best optimal solutions.

(5) ~~Education~~ execution.

It execute the best optimal sol from the searching algorithum to reach the goal state from the current state

Example :- 8 puzzle problem :

① Initial state :-

1	8	2
3		7
4	5	6

Goal state

1	2	3
4	5	6
7	8	

② Actions
upward, downward, left, Right

upward

1	8	2
3	↑	7
4	5	6

downward

1	8	2
3	↓	7
4	5	6

left

1	8	2
3	←	7
4	5	6

right

1	8	2
3 →		7
4	5	6

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(3) Transitions model

1	8 ←	2
3	↓	7
4	5	6

1	2	
3	8	7
4	5	6

(4) goal tat (defination)

(5) Path cost (own define)

VLSI (chip designs) google (TSP) algorithm
very large scale integration

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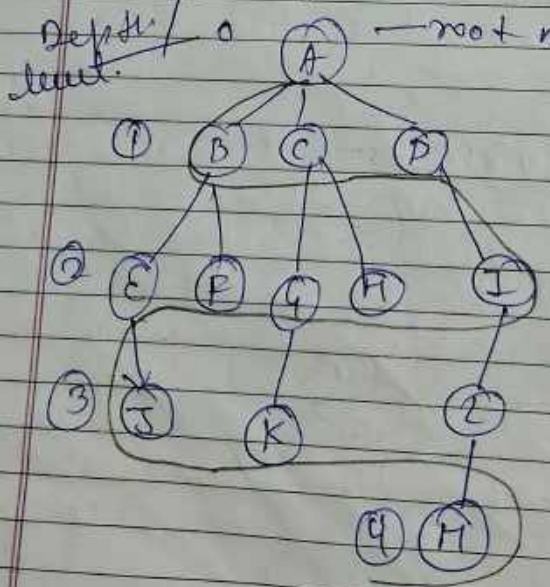
Searching techniques

- Uninformed (blind)
  - Breadth first search (BFS)
  - Depth first search (DFS)
- Informed / Heuristic (additional inf)
  - Best first search / greedy search
  - A* search
  - A^cr* "

Breadth first Search

Depth / level 0 — root node



A B C D I H G F E
J K L M

→ check A is goal state or not, check every node is goal state or not.

→ BFS use queue D.S.

→ # FIFO (Queue)

→ It search level by level

A to B to C to D

comes D child node

← [A]

so, child node comes of A

← [B C D] ←	← [H I J K] ←
← [C D E F] ←	← [I J K] ←
← [D E F G H] ←	← [J K L] ←
← [E F G H I] ←	← [K L] ←
← [F G H I J] ←	← [L] ←
← [G H I J] ←	← [M] ←
← [H I J] ←	

A B C D E F G H I J K L M

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It search breadth node first then you go depth wise.

Q. To find L.

A B I L
root → find L

1. compute technique
2. find a optimal sol. (shortest part from this techniques)
3. It will search the shallowest (neighbourhood) node first, then its goes deepest node

M.Imp

4. Time & space complexicity.

$O(V + E)$
vertices (no. of nodes) — edges

(DS)

→ Time & space comp.

for A I

$O(b)^d$ (fix)

b – branch factor
d – depth

b = 3 maximum child nodes
d = 4 depth.

$O(3^4) = 3 \times 3 \times 3 \times 3$

$O(81)$

for find I (T.C)

b = 3
d = 2
$O(3^2) = O(9)$

~~Informed~~ You explain problem 8 puzzle problem with both search

~~(8 puzzle problem)~~

Example) Puzzle problem with on Informed search.

(1)

1	2	3
	4	6
7	5	8

(initial state)

1	2	3
4	5	6
7	8	

(final state)

Define

(i) Initial state

(ii) Actions. (up, down, left, right)

*. Actions/directions are called child nodes for each element/node.

Q. solve with the state space (total no of state for 3×3 g.s)

1	2	3
	4	6
7	5	8

(vacant tile move)

U R D

	2	3
1	4	6
7	5	8

1	2	3
4		6
7	5	8

1	2	3
7	4	6
	5	8

(shallowest move) 1

R D

2		3
1	4	6
7	5	8

1	2	3
	4	6
7	5	8

D U L R

1	2	3
4	5	6
7		8

1		3
4	2	6
7	5	8

1	2	3
	4	6
7	5	8

1	2	3
4		6
7	5	8

2

3

R.

1	2	3
4	5	6
7	8	

(goal state find)

Advantage

① Solution will find (w/. its a goal state)

② optimal solution

Time Complexicity

① It will take more time

$O(b)^d$

$d = 3$, $b = 4$.

$b \rightarrow (4)^3 = 64$

0	1	0
1		1
0	1	0

1	2	3
4	5	6
7	8	

9 space.

$\frac{4\times2 + 4\times12 + 1\times4}{9}$

$= 8 + 12 + 4 = 24$

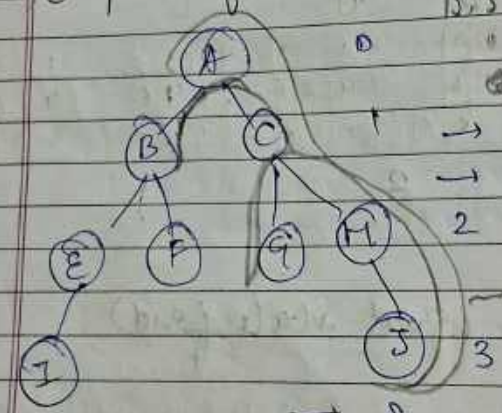
$$\frac{8 + 12 + 4}{9}$$

$$= \frac{24}{9} = 2.6$$

$$(3)^3 = 27$$

Depth first search (DFS)

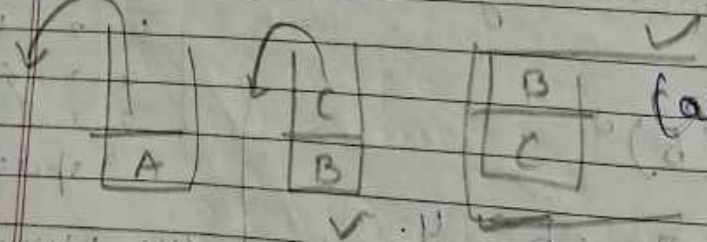
D.S use (stack)
(LIFO)
last in first out



→ LIFO
→ Backtracking

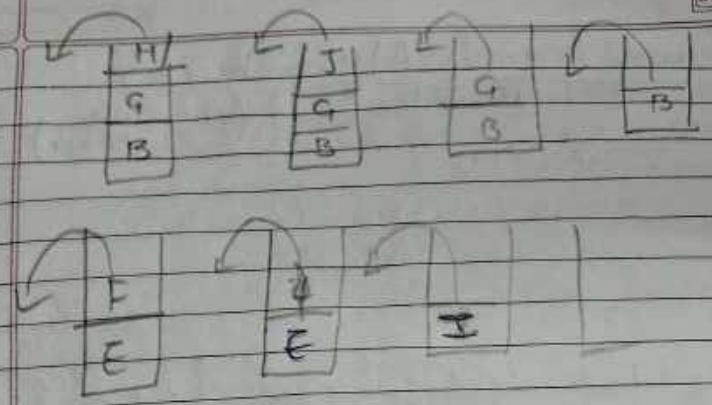
~ Backtracking

→ Search the deepest node first. then move to the shallowest node.



(another case)

(Alphabetically)



A C H J G B F E I

Disadvantage advar

→ Incomplete technique

→ if search space is infinite then it will give no o/p.

T.C $O(b)^d$

$b = 2$

depth $d = 3$

$(2)^3 = 8$

(no searches)

★ Informed Search Algorithm (for time bachana)

Heuristic → Information → h(n) (function)

① calculate linear distance
Euclidian distance

(x_1, y_1) (x_2, y_2)

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

radom

② Manhattan distance ("fast algorithm")

(short path find) vertical, horizontal distance of two points.

8 Puzzle problem with heuristic

1	2	3
	4	6
7	5	8

(F)

1	2	3
4	5	6
7	8	

(G)

no. of misplace child (5)
for all elements :-

$$0 + 0 + 0 + \overset{5}{1} + \overset{4}{1} + 0 + 0 + \overset{5}{1} + \overset{8}{1}$$

only 1 vacant count

2 vacant mean pick

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1	2	3
	4	6
7	5	8

blank space place

R U D

1	2	3
4		6
7	5	8

h(2)

1+1

	2	3
1	4	6
7	5	8

h(4)

1+1+1+1

1	2	3
7	4	6
	5	8

h(4)

1+1+1+1

we follow this

D U L R

1	2	3
4	5	6
7		8

h(1)

1		3
4	2	6
7	5	8

h(3)

1	2	3
	4	6
7	5	8

h(3)

1	2	3
4	6	
7	5	8

h(3)

R L U

1	2	3
4	5	6
7	8	

1	2	3
4	5	8
	7	8

1	2	3
4		6
7	5	8

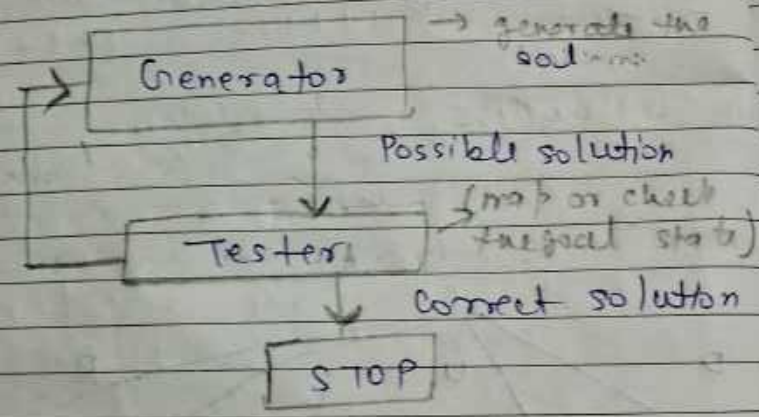
the find our goal state

Informed Search

1. Generate & Test | British ~~[illegible]~~ search (museum search)
2. Best first Search | Greedy Adm Possible Alg
3. A* Algorithum
4. AO* Algorithum

(easiest one)

Generate and Test Algorithum



→ generate no. of soln with heuristic and find soln in each case.

Steps:-

1. Generate a possible solution.
2. Test to see if this is a actual solution.
3. If a solution is found, the current

quit, otherwise go to step 1.

Drawback :- T.C ↑, but you will def get optimal soln.

generator

① It will have to generate all the possible sol — (generator)

② It should not produce / ~~[illegible]~~ generate the non-random soln / repeadetly soln

Best first Search (min distance)

→ It is greedy to the lowest heuristic value.

STEPS.

① let ('OPEN') be a priority queue containing initial state.

② If ('OPEN') is empty, then return failure.

③ Now, name this ('OPEN') a node and then remove first ('OPEN')

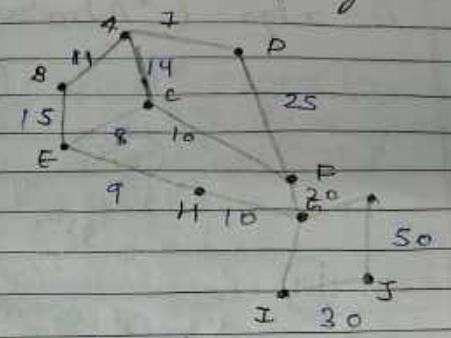
open the — initial node.
If it does not have child node (failure)

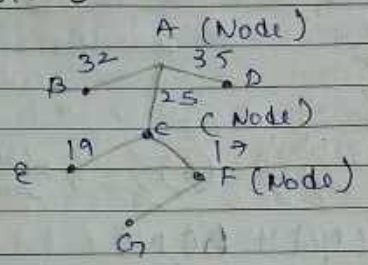
4. If node is a goal then return the path from initial to node.

5. ~~As~~ else generate all successor of node and put the newly generated node in to ('OPEN') according to their heuristic function.

Best first Search (Heuristic) (Goedy Search)

A ⟶ G = 40
B ⟶ G = 32
C ⟶ G = 25
D ⟶ G = 35
E ⟶ G = 19
F ⟶ G = 17
H ⟶ G = 10
G ⟶ G = 0

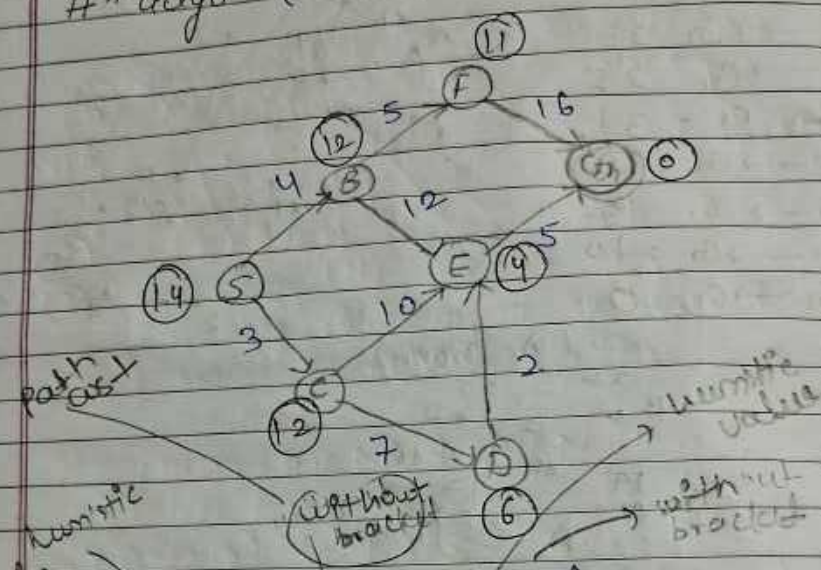




Key points

(i) It will always give you good solution.
(ii) No guarantee for the optimal solution.
(iii) It is always goedy for heuristic function.

A* algo (*Admissible)



$f(N) = g(N) + h(N)$

$g(N)$ = Actual cost from start node to n node

$h(N)$ = Estimate cost " n node to Goal node

$S \rightarrow B$	$S \rightarrow C$
$f(S) = 4 + 12 = 16$	$f(S) = 3 + 11 = 14$
$SC \rightarrow D$	$SC \rightarrow E$
$f(SC)\ 3 + 7 + 6 = 16$	$f(SC) = 3 + 10 + 4 = 17$
$SCD \rightarrow E$	$SCDE - G$
$3 + 7 + 2 + 4 = 16$	$3 + 7 + 2 + 5 + 0 = 17$

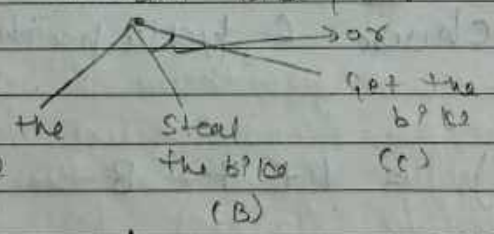
AO* Algorithum

- It is a variant of A* algorithum which is designed to be more flexible and capable of adapting the changing environment.

- The AO* algorithum divide any given difficult problem into a smaller group of problem that are resolved using AND — OR graph.
  - AND → (+)
  - OR → (either this or this)

examples

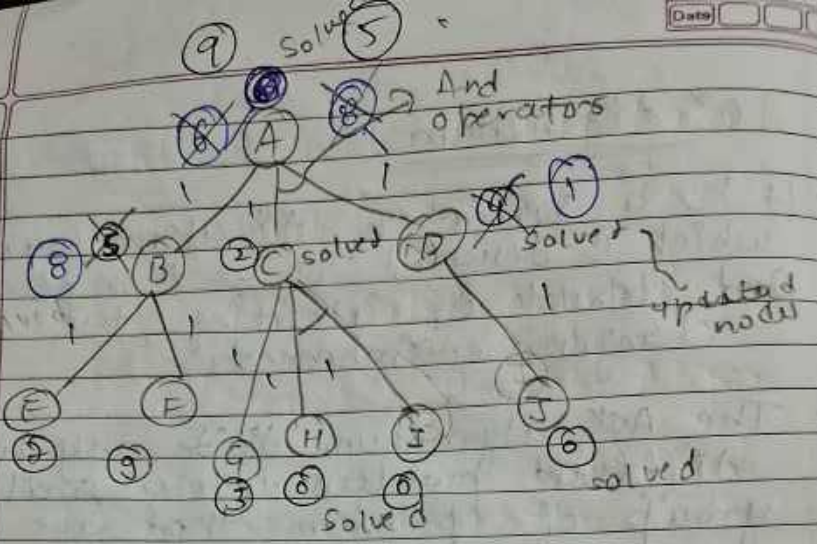
I want bike | car



Buy the bike (or) steal the bike (and) Get the bike

other examples :-

1. exam
2. games



$f(N) = g(N) + h(N)$

$f(A \to B) = 1 + 5 = 6$

$f(A \to C + D) = (1+2) + (1+4)$

$3 + 5 = 8$

we choose 6 wala path

$f(B \to E) = 1 + 7 = 8$

$f(B \to F) = 1 + 9 = 10$

Take A Right path now 8 wala :-

$f(C \to G) = 1 + 3 = 4$

$\cancel{f(C \to H) = 1 + 0 = 1}$

$\cancel{f(C \to F) = 1 + 0 = 1}$

$F(C \to H + I) = 1 + 0 + 1 + 0 = 2$

C is already 2 (no change) (no updation)

$F(D \to J) = 0 + 1 = 1$

D is - 4 # I (update)

S is correct path.

The main focus of this algorithum is that we have to find firstly level 1 heuristic value and then level 2 ~~heuristic value~~ after that update the values with going upward means towards the root node.

(local domain
about next step know
know neighbour area!

local Search Algorithum

1. Hill climbing Algorithum
2. Stimulated Annealing
3. Genetic Search Algo.

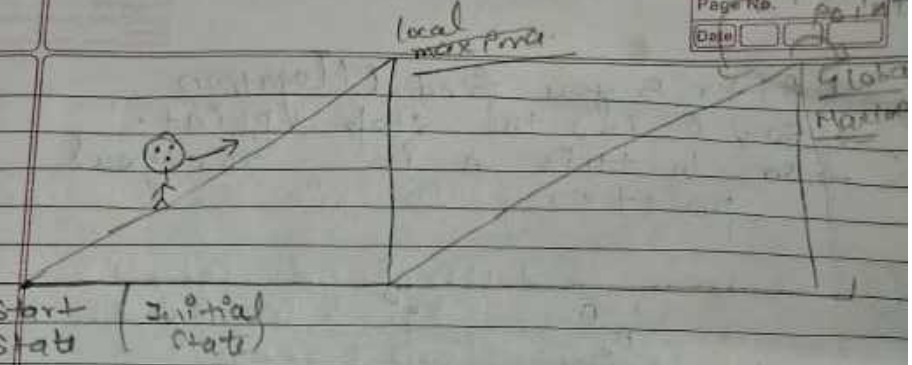
{main goal is the goal} does not affect of path.

→ it only and store the goal that it is followed.

Hill climbing Algo.

→ Hill climbing Algorithum is a local search Algorithum which continuously moves in the direc of increasing values to find peak of the mountain or best solution to the problem.

→ only knowledge about local domain



① local maxima, global maxima

Approach - next state is better than the current ".

→ If the next state is not better so the algo will stop

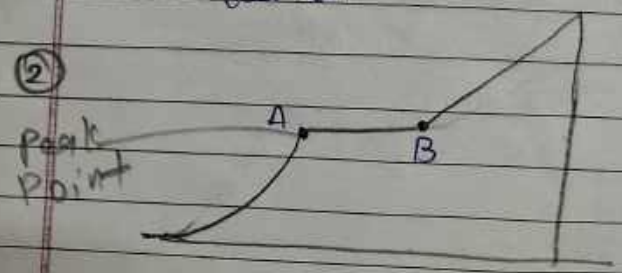
You can explain

Drawback

① local maxima
② flat surface / plateau
③ Ridge
④ You can't change your direction

→ good solution not the optimal solution.

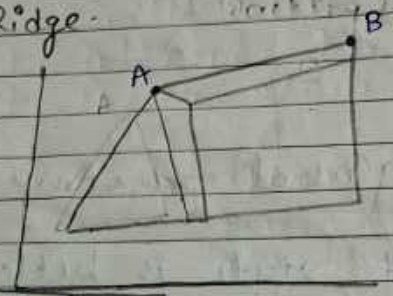
②



Current state = next state

→ After B you find allogation
→ So, B is the stop point.
→ So, In this A is the goal point.

③ Ridge.



Key points :-

① memory efficient :- It is memory efficie

② It is used for problems ~~for~~ with multipu solution.

Matlergy
↳ Metals knowledge

matelogy process

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Stimulated Annealing

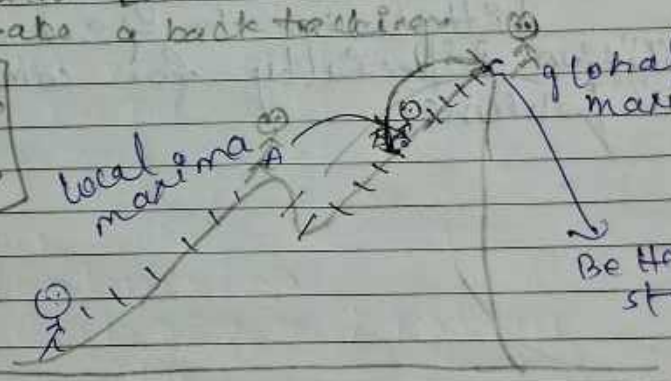
Backtracking allowed
Find fit into solution entity

Annealing :- Check the environment

In matelogy Annealing is a process to heating a metal an excessive temperature after which gradually cooling it to lessen defect and optimize its crystalline structure.

① Its also check the other side and take a back tracking.

Draw diagram after every step



Check all the environment when its find the (elimated / best step) so its directly jump in the step.

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key points

1. It is easy to Code for complex problems.
2. It will give good solutions.
3. It allows backtracking.
4. It is a slow process.
   for exploring state and environment so it will take time.
5. It may require a large no. of itteration to find a good solution specially for complex problems.